

Phase 3 - A Tracked Dynamic Patient State for PatientSim

Github URL

<https://github.com/donghalim7/PatientSim> (Phase 3 code: src/phase3/, src/prompts/simulation/phase3_state_protocol.txt; results: phase3/)

Project Motivation & Goal

We continue the Phase 2 goal: make PatientSim's patient turn-aware so that, as an ED interview proceeds, the patient's state evolves rather than staying static. Phase 1/2 showed PatientSim is a content-disclosure simulator with a static state - it reveals profile facts when asked, but tone, focus, and care-seeking do not change across turns (the same focal question asked early or late yields a near-identical answer). Phase 2's prompt-only intervention (V17) unlocked turn-by-turn drift on acute cases but left subacute cases (pneumonia) flat, and a static demonstration prefix carries no patient state that actually persists or updates.

Phase 3's goal is therefore to implement that dynamic patient state for real: an explicit, tracked felt state that updates every turn so the patient is (1) turn-aware, (2) sensitive to the doctor's interview strategy (drift when the chief complaint is delayed; little drift when it is addressed early), and (3) clinically faithful - never inventing symptoms, and for slow-progression (subacute) cases keeping clinical severity flat while only worry/engagement may move. The target is to beat the V17 prompt-only ceiling on the Phase 2 evaluator without sacrificing safety.

Method

We add DynamicPatientAgent, a subclass of the original PatientAgent (the original agent and the shared model layer are unmodified). It carries a felt state of three 1-5 variables: clinical_severity (how intense symptoms feel), interactional_engagement (focus / clarity), and care_seeking_pressure (expressed worry).

Hybrid policy, not model self-update. We first let the model emit its own next state each turn; this failed empirically - at temperature=0 it left the state flat, it keyed drift to focal questions instead of delay (scoring higher under high-yield than low-yield, the wrong direction), and once raised its score by inventing symptoms (faithfulness dropped). So Phase 3 uses a hybrid: Python owns the state trajectory through an explicit clinical policy; the model only renders words expressing the injected state.

Per turn: (1) a policy sets the target state as a function of (case acuity, doctor strategy, turn), clamped to a gradual +/-1 step; (2) the state plus a short phrasing note derived from it ("voice some concern", "answer a bit more briefly") plus the doctor question are injected; (3) the model returns only the patient's line, and the state is logged to a state_trace. The policy ramps from an initial state to a per-condition target:

Condition			end state (sev/eng/care)	onset	rationale
acute, delayed	focal	(low-yield)	3 / 3 / 3	0.33	symptom + affective drift
subacute, delayed	focal	(low-yield)	2 / 4 / 3	0.45	severity flat, affective drift only
acute, focal early (high-yield)			2 / 5 / 1	-	attended to -> no drift
subacute, focal	early	(high-yield)	2 / 5 / 2	0.55	mild concern only

The key choice is that subacute severity is held flat: a waiting patient may grow more worried or less engaged, but the disease does not "worsen" in minutes, so no new symptoms are produced. We also reuse V17's two prompt edits (relax the consistency rule, remove the one-sentence cap) so the felt state can surface. No model, temperature, or persona change.

Evaluation Strategy

We reuse the Phase 2 five-dimension LLM-as-judge unchanged: three dynamic dimensions (turn-to-turn state drift, interactional engagement, care-seeking pressure) and two safety dimensions (state-drift plausibility, clinical-fact faithfulness), each 1-5, combined into a weighted overall. For each of the 5 cases (3 acute: cerebral infarction, MI, intestinal obstruction; 2 subacute: pneumonia 62M, pneumonia 92F) we compare Baseline (unmodified) -> V17 -> Phase 3 under the low-yield (delayed-focal) script, controlled persona (CEFR B / plain / high / normal), backend gemini-2.5-flash. Success requires: (i) match/beat V17 on acute, (ii) lift subacute above its flat baseline, and crucially (iii) without dropping plausibility or faithfulness - a higher score from invented deterioration is a failure. We also score the high-yield script (a correct dynamic patient should drift less when the complaint is addressed early). Because the judge is non-deterministic, headline numbers are mean +/- std over 3 seeds (42/7/13, temp 0.7). During Phase 3 we found the judge's profile view omitted medication and living_situation, wrongly flagging profile-grounded answers as "invented"; we added those fields and re-scored V17 with the fixed judge for fairness.

Results and Discussion

Headline (low-yield; Phase 3 = N=3 mean +/- std):

Case	Baseline	V17	Phase 3	drift	plaus	faith
cerebral (acute)	2.20	3.70	4.37 +/- 0.09	4.0	5	5
MI (acute)	2.20	2.60	4.30 +/- 0.00	4.0	5	5

Case	Baseline	V17	Phase 3	drift	plaus	faith
intestinal (acute)	~2.2*	4.30	4.30 +/- 0.00	4.0	5	5
pneumonia (subacute)	2.05	2.20	4.23 +/- 0.09	4.0	5	5
pneumonia 92F (subacute)	1.90**	2.20	4.23 +/- 0.09	4.0	5	5

No measured unmodified run for intestinal (~2.2 is an inferred static reference). *1.90 is V17 on 92F (behaviourally static), scored under the pre-fix judge; no separate unmodified run exists. Measured unmodified baselines exist for cerebral/MI/pneumonia-62M.

Phase 3 matches or beats V17 on every case, with overall std <= 0.09 and plausibility = faithfulness = 5.0 across all seeds - the gains are stable and safe.

Strategy sensitivity (single sample): acute cases drift under delayed questioning but fall back to a flat ~2.2 when focal questions come early (cerebral 4.60->2.20, MI 4.30->2.20, intestinal 4.30->2.20), reproducing - under explicit control - the doctor-strategy dependence Phase 2 observed.

Qualitative (pneumonia, subacute, low-yield): sev stays 2 throughout; only worry/engagement move, with no symptom outside the profile: T1 [sev2 eng5 care1] "I have shortness of breath." -> T5 [sev2 eng4 care2] "I use Spiriva and Advair. Is my breathing okay?" -> T6 [sev2 eng4 care3] "...The shortness of breath started yesterday. I'm worried."

Do the metrics capture what we claim? Yes for the limitation and the acute gains: baseline is a uniform ~2.0-2.2 with drift=eng=care=1 and safety=5 - i.e. the judge confirms a static but faithful patient, exactly the gap we targeted; on acute cases Phase 3's drift reflects genuine, profile-grounded symptom progression plus rising worry, with safety intact. One honest caveat on the subacute result: the lift (e.g. pneumonia 2.05->4.23) is real and faithful, but it is affective drift only (rising worry / falling engagement) with severity held flat. The evaluator's single drift dimension does not separate clinical-progression drift from affective drift (and overlaps care-seeking), so a purely-affective subacute case scores the same drift (4.0) as an acute case that genuinely worsens - making subacute (4.23) look nearly equal to acute (4.30). They are equal in score but different in kind. We therefore do not claim subacute patients are "as dynamic as" acute ones; we claim Phase 3 makes them dynamic on the affective axis in a way that is plausible and faithful for a slow presentation - which is the clinically appropriate behaviour, and which the current metric rewards somewhat too generously. Limitations: the judge is noisy (engagement most fragile - hence N=3); the policy is hand-designed with hand-labeled acuity (reliable and faithful, but drift shape is fixed not emergent); single backend/persona.

Future Work

The most important next step for patient simulation is a patient state that the simulator infers and updates from the live dialogue (e.g. reassurance lowering worry) rather than scripting it - while staying provably faithful, e.g. by letting the model propose state nudges within a plausibility envelope derived automatically from the profile instead of hand-labeled acuity. In parallel, the evaluator should separate clinical-progression drift from affective drift, since collapsing them obscures whether a "dynamic" patient is faithfully worried or implausibly deteriorating.